

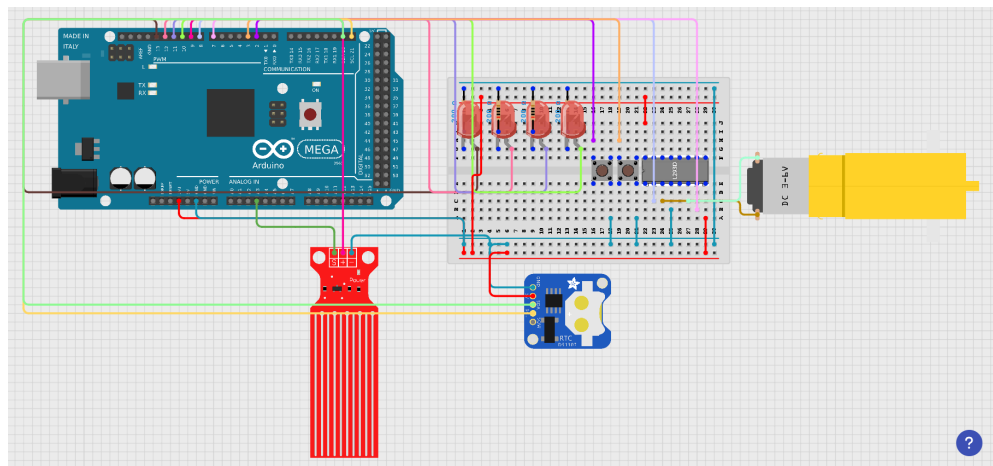
Final Project

1. Project Motivation:

This project is inspired by the numerous times that I have discovered that my dogs' water bowls are all low, because they are located in out-of-the-way areas that it render them easy to overlook by simply not walking by them, and thinking that it would be much better if I had a system that would notify me when the water levels reached a certain point. So, presumably, it would be used by people in similar situations to mine - having pets that they wanted to ensure had ready access to fresh water, which was located in places that it might be easy to allow it to run low during particularly warm times of the year or after significant exercise, and who would like some form of system in place to notify them when a refill was necessary. An embedded system is a reasonable solution because this involves two very simple functions - monitoring the level of water in an area, and activating a notification system - in real time, without many parts needed to run the system.

2. Proposed System Overview:

a. System Diagram:



i.

b. System Component List:

- i. Arduino MEGA 2560 connected to computer with Arduino IDE x1
- ii. LED x4
- iii. 330 Resistors x4
- iv. DIP Push Button x2
- v. RTC 1307 Module x1
- vi. 3-6V Motor x1
- vii. Fan Blade x1
- viii. Solderless Breadboard x1
- ix. Jumper Kit

3. Expected System Behavior:

While the system is in its OFF state, it turns on a single LED to reflect that it is OFF, and otherwise turns off all other output devices in the system. While the system is in its IDLE state, it turns on an LED reflecting this while turning off all other LEDs, and displays the currently detected water level. In this state, a button may be used to decide whether it will enter into an LED-only alert state if the water level drops below a certain point, or an LED-and-audio alert state if the water level drops below a certain point. Once the water returns to an appropriate level, the system returns to an IDLE state. At any point while on, if any of the system's expected components are disconnected, it enters into an ERROR state. Furthermore, at any point while on, the user may return the system to an OFF state by pressing a button.

a. Module Design

This system is notionally divided into six modules - a control module, a sensor module, a user input module, a display module, a 'screen' module, and a communication module.

The control module primarily consists of the microcontroller, and will be responsible for the 'decision making' of the system. It will contain all the programming needed to ensure that the system transitions between states properly in response to various signals, and behaves appropriately in those states.

The sensor module consists of only the water level sensor. This module is responsible for acquiring the information that will, barring errors, trigger changes between active and inactive states in the system.

The user input module consists of a pair of buttons. It will be responsible for accepting inputs from the end user to control the desired behavior of the system.

The display module consists of a set of LEDs and a fan. This will provide basic information to the end user and notify them of state changes.

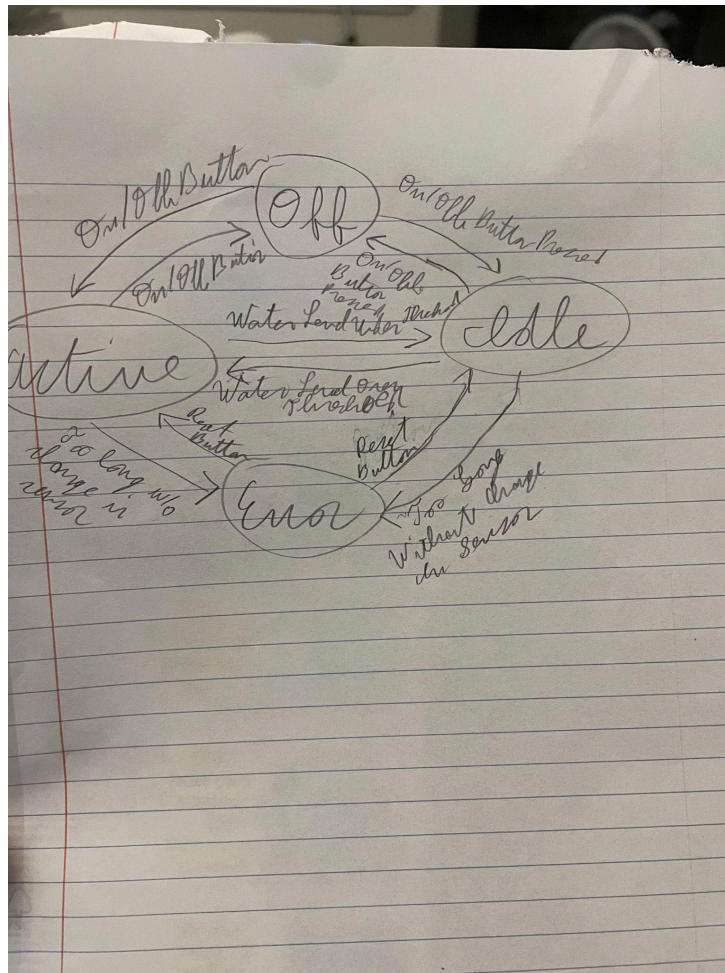
The screen module consists of a computer screen. It will display reports as to the history of the system and movements between states and such.

The communication module will handle the transmission of information between the other modules. It will accept inputs from the sensor and user input modules, ensure they are appropriately transmitted to the control module, and relay instructions from the control modules to the other modules in order to ensure that the system behaves as intended.

According to this design, the communication module is the hub of interaction between all other modules. It will handle the sorting and

transmission of signals, allowing the system to function as a cohesive whole, and permitting each module to 'focus' on only its designated task.

b. Control Logic State Diagram



c. Control Logic State Descriptions

i. OFF State

Purpose: System is powered but not operating..

Behavior:

- No sensor reading.
- Actuator is OFF.
- Only ON button interrupt is functional.
- OFF LED in ON state.

Transition:

- ON button (interrupt) -> IDLE

ii. IDLE State

Purpose: System monitors water level in container.

Behavior:

- Reads water sensor data.
- Displays sensor value on computer screen via serial monitor.

- Logs readings periodically with time-stamp.
- Actuator OFF.
- IDLE LED in ON state.

Transition:

- Water level sensor detects too low level -> ACTIVE
- Water level sensor does not detect a change for significant period of time (five minutes for demonstration purposes) -> ERROR
- OFF Button Pressed -> OFF

iii. ACTIVE State

Purpose: System visually indicates that water level is below expected threshold and activates fan.

Behavior:

- Reads water sensor data.
- Screen displays a low water level notification and log of recorded levels.
- Logs state time at which entered state with timestamp.
- ACTIVE LED in ON state.
- Actuator ON.

Transition:

- Water level is placed above expected range -> IDLE
- Water level sensor does not detect a change for significant period of time (five minutes for demonstration purposes) -> ERROR
- OFF Button Pressed -> OFF

iv. ERROR State

Purpose: Detect malfunction and enter safe mode.

Trigger Conditions:

- System does not detect a change in water level for significant period of time.

Behavior:

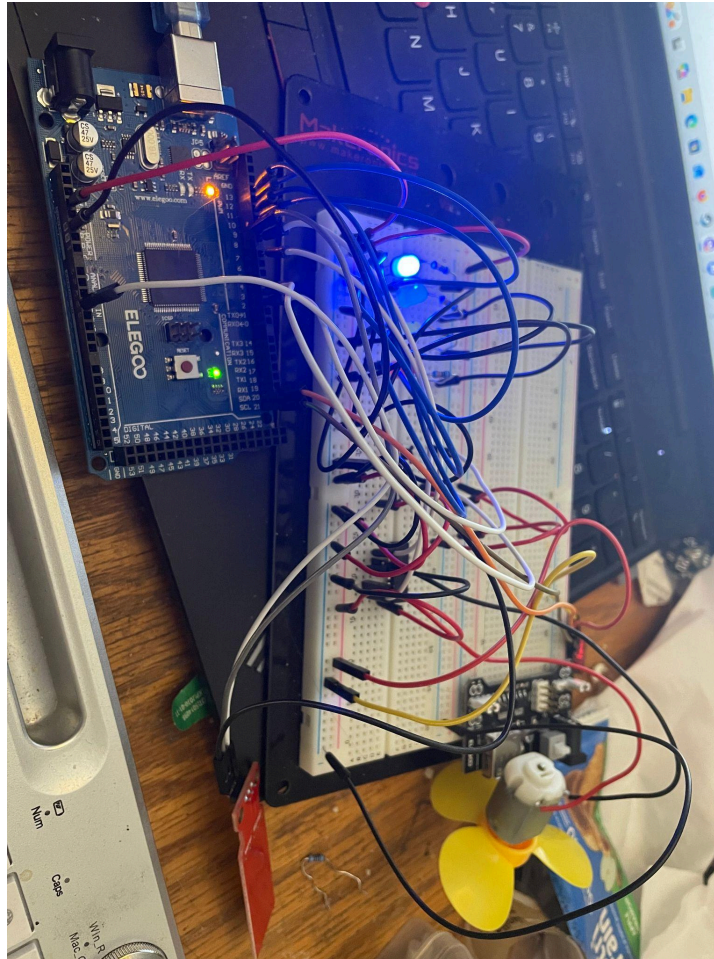
- Actuator OFF.
- ERROR LED in ON state.
- Computer screen displays error message in serial monitor.
- Serial log entry with time stamp of time that entered state.
- System stays in ERROR.

Transition:

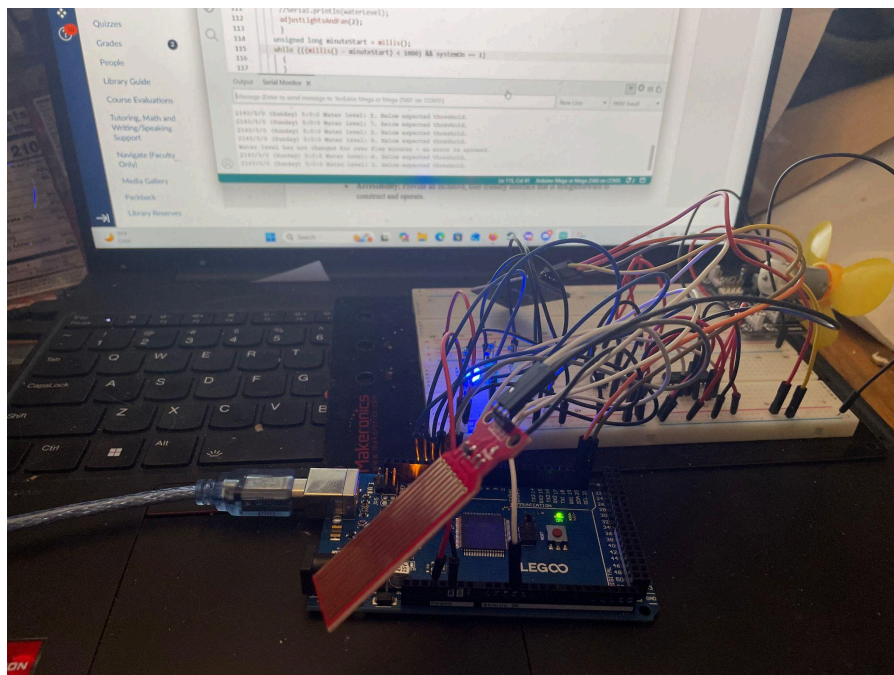
- RESET button is pressed -> IDLE
- OFF button is pressed -> OFF

4. Results

a. Full Circuit Image:

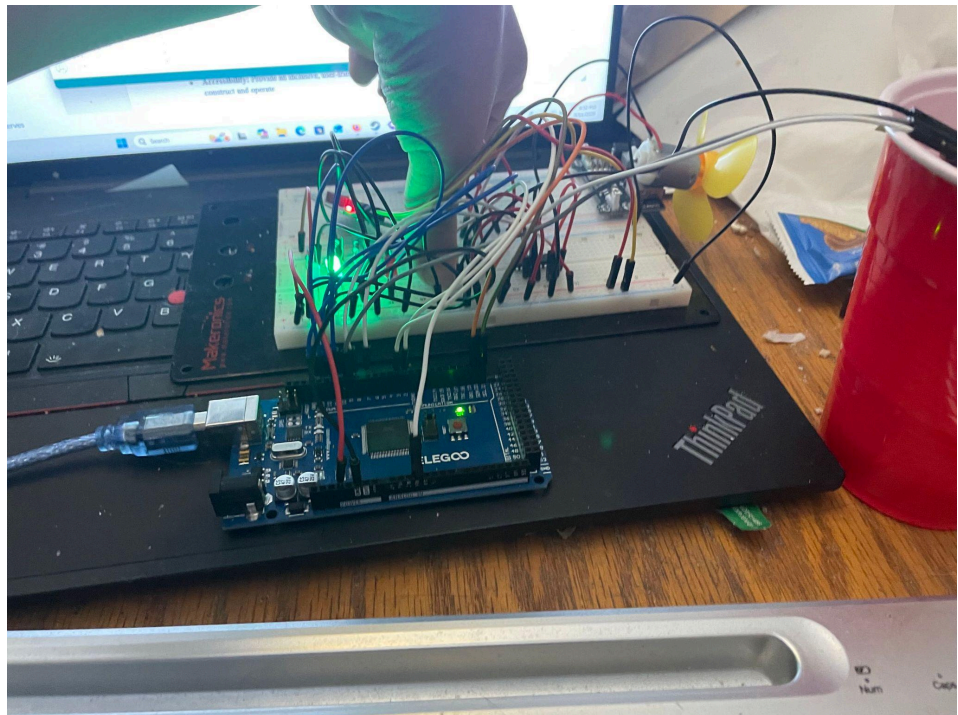


- i.
- b. OFF Stage



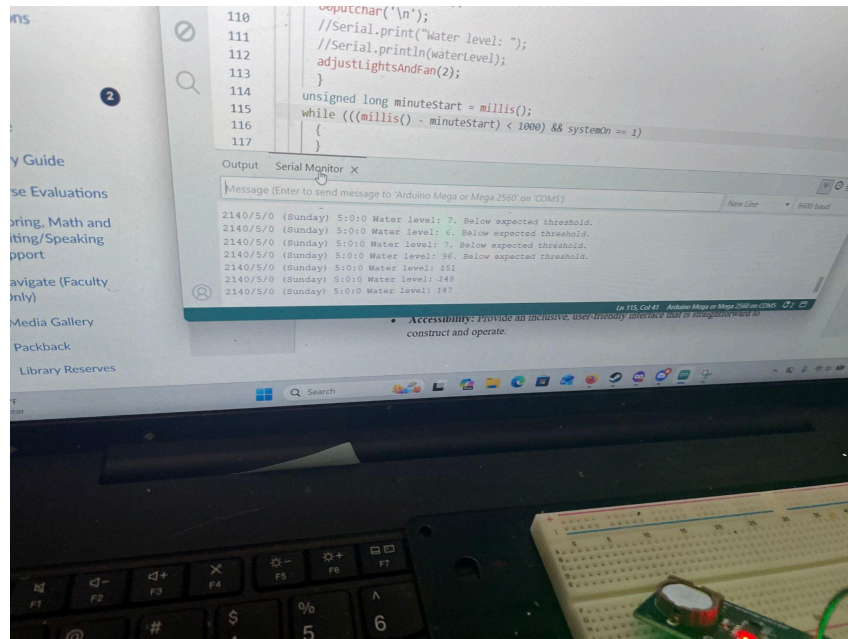
i.

c. IDLE Stage Circuit



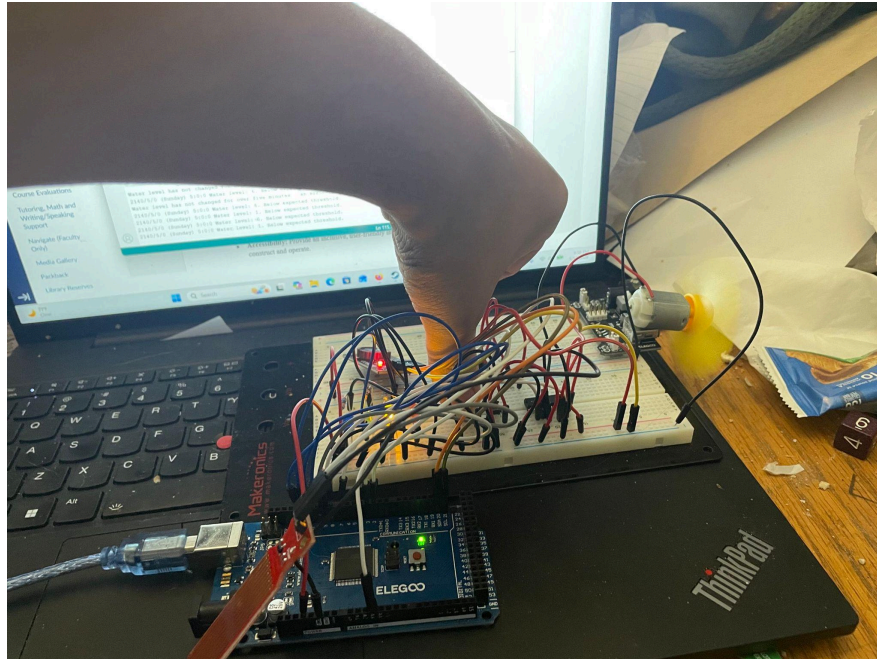
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d. IDLE Stage Serial Monitor



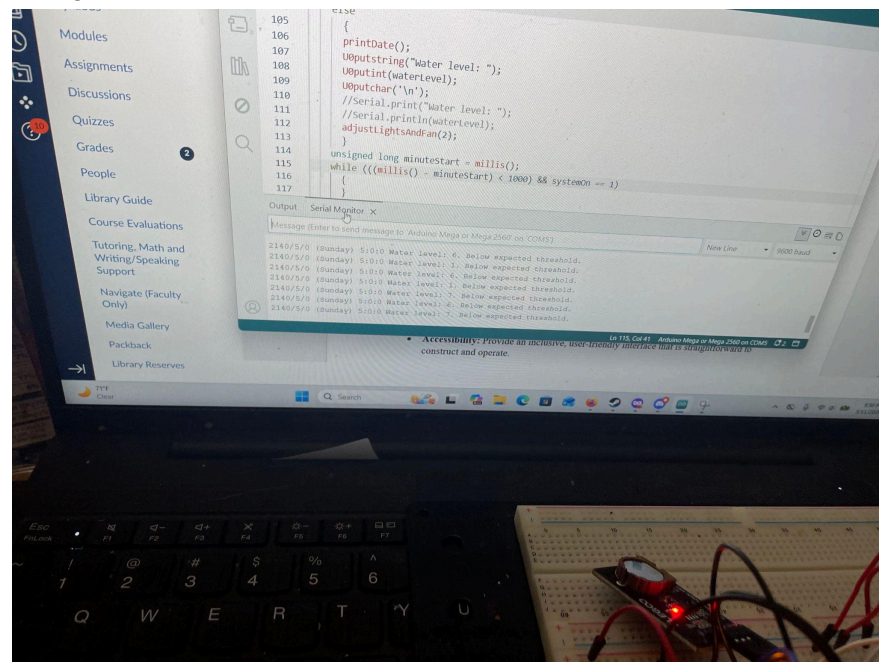
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e. ACTIVE Stage Circuit



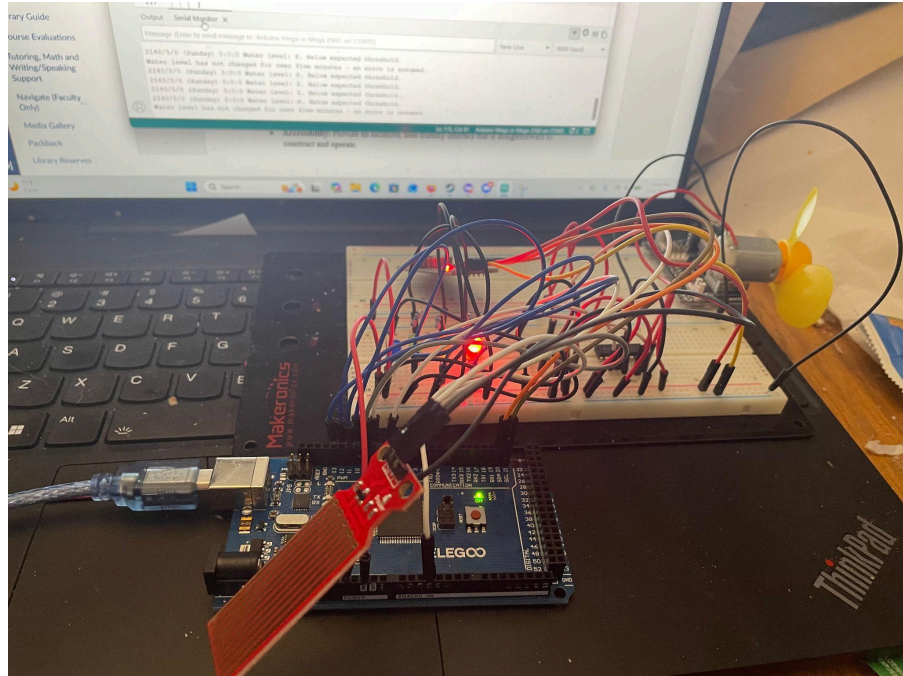
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f. ACTIVE Stage Serial Monitor



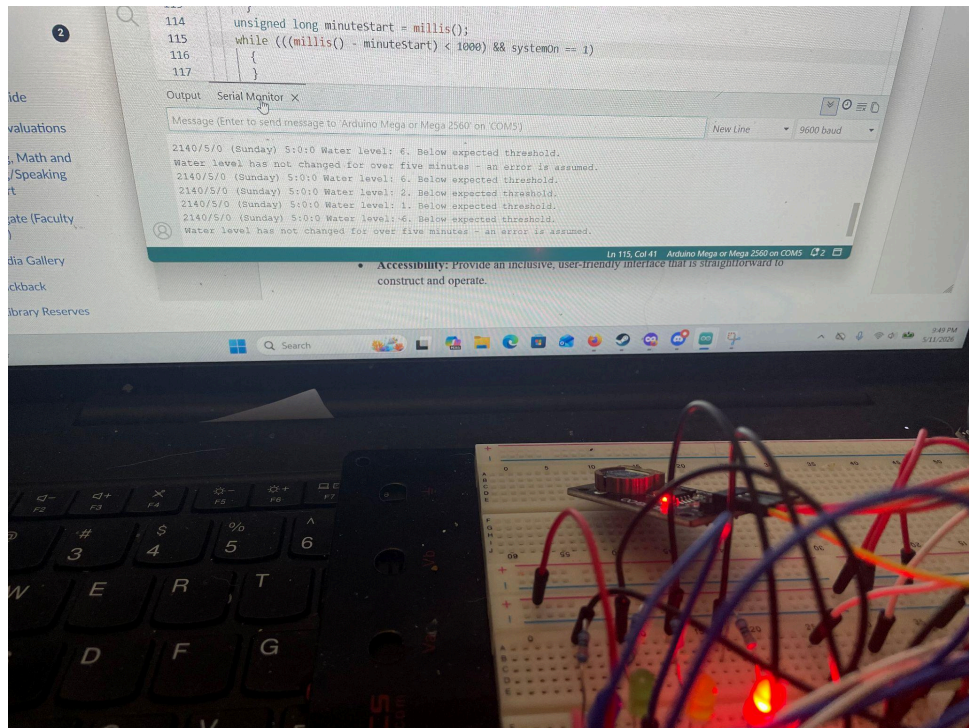
i.

g. ERROR Stage Circuit



i.

h. ERROR Stage Serial Monitor



i.

5. Environmental & Energy Impact:

The project has been simplified from prior iterations, with the net result being a mostly lateral move in energy consumption. The project has a single active state that it may enter, and thus does not have a less-consumptive mode that it may enter, but the buzzer has been removed from the design, reducing the energy consumption levels. The 'error' state's current

time frame would significantly reduce the amount of time the project might spend in either its idle or active states, thus reducing its power draw, but in any version that had been adjusted for non-demonstrative use, this factor would likely no longer apply. All that said, the components required by the system are simple, already mass-produced and require relatively little power to function, so this system's impact, both in terms of energy and raw material usage, would be very light, and while functioning, the system would do little to nothing to contaminate the environment in any way. While a refined version of the system could likely further reduce its power requirements, as it stands, there is, I believe, no reason to expect any significant impact if a hypothetical mass-market version that significantly resembled this 'prototype' were developed.

6. Conclusions

The system only approximately functioned as intended. Due to something that was determined to be a wiring error, but whose cause could not be found, the system developed a quirk where in the interrupts for both buttons triggered constantly. This caused the system to randomly turn off for seemingly no reason. However, that aside, it appeared to generally function as intended.

That being said, the system has significant limitations. Most relevantly, attempting to use this to actually measure the level of water in a container over a long period of time is physically infeasible. Doing so requires exposing significant amounts of delicate electronic equipment, of most notable mention being the computer acting as the power source in the project's design, to the potential of being destroyed by water damage. In addition, as demonstrated by the video, it is difficult to position the fan such that it is not prevented from rotating, which, I anticipate, would cause damage to the fan in the long term.

In addition, the frequency of readings, and the length of time that water levels could go unchanged before entering into an error state, would need to be adjusted. As mentioned in the video, they are currently of a length to fit project constraints and provide a viable filming window, but these time frames are far too short for the project's intended purpose.